Unit 2.2 Printables: Inherited Instructions

Every table students only copy or fill in is printed here, so they do not spend class time drawing it. Cut each piece on the dashed line and glue it into the matching “Glue in” box on that day's notebook setup page (Unit22_Student_Notebook.pdf). Print on white paper, black ink only. Answers: Unit22_Printables_Teacher_Key.pdf.

Day	Printable	Use	Copies
Day 1	Puppy litter data	Glue in · write on it	1 per student
Day 2	Card table: 3 offspring	Glue in · write on it	1 per student
Day 4	Genotype to phenotype: pea plants	Glue in · write on it	1 per student
Day 4	Punnett squares: steps and worked example	Glue in · copy of the board	1 per student
Day 5	Monster Genetics: the coin rule and Table 2	Glue in · reference and write on it	1 per student
Day 6	Punnett square: skin color and eyes	Glue in · write on it	1 per student
Day 6	Prediction vs. class count	Glue in · reference	1 per student
Day 7	UV light and mutations	Glue in · reference	1 per student
Day 7	Who passes it on? Five cases	Glue in · write on it	1 per student
Day 8	Case data: three cases	Glue in · reference	1 per student
Day 9	The new coat pattern: two kennels	Glue in · reference	1 per student
Day 10	My target tracker	Glue in · write on it	1 per student
Day 10	Hydrangea soil data	Glue in · reference	1 per student

Days 3 and 11 need nothing glued in (Day 3 notes are copied into the notebook; Day 11 is the CSA).

UNIT 2.2 · DAY 1 · GLUE IN · WRITE ON IT

Puppy litter data

Next to each puppy trait, write M if it matches Mom or D if it matches Dad. Puppy A is done for you.

	Fur color	Ear shape	Pattern
Mom	Black	Floppy	Spotted
Dad	Brown	Pointed	Solid
Puppy A	Black (M)	Floppy (M)	Solid (D)
Puppy B	Brown	Pointed	Spotted
Puppy C	Black	Pointed	Spotted
Puppy D	Brown	Floppy	Solid

UNIT 2.2 · DAY 2 · GLUE IN · WRITE ON IT

Card table: 3 offspring

Draw one card from each of the six piles. Write what you drew, then put the cards back. Offspring 1 is done for you.

Offspring	Fur color (P1 + P2)	Ear shape (P1 + P2)	Pattern (P1 + P2)
1 (done for you)	Black + White	Floppy + Pointed	Spotted + Solid
2			
3			

UNIT 2.2 · DAY 4 · GLUE IN · WRITE ON IT

Genotype to phenotype: pea plants

T (tall) is dominant to t (short). TT is done for you. Fill in the phenotype for Tt and tt.

Genotype	Phenotype
TT	Tall (done for you)
Tt	
tt	

UNIT 2.2 · DAY 4 · GLUE IN · COPY OF THE BOARD

Punnett squares: steps and worked example

1. Write one parent's two alleles across the top. **2.** Write the other parent's two alleles down the side. **3.** Fill each box with the allele from its column and the allele from its row.

Worked example: Tt × Tt

	T	**t**
T	TT	Tt
t	Tt	tt

Result: TT, Tt, Tt, tt. That is **3 tall to 1 short**.

Each box in a Punnett square shows one possible ____ of alleles.

UNIT 2.2 · DAY 5 · GLUE IN · REFERENCE AND WRITE ON IT

Monster Genetics: the coin rule and Table 2

Flip a coin twice for each trait. Heads = Allele 1 (capital letter). Tails = Allele 2 (lowercase). Example, Eyes: heads, tails gives **Ee**. E is dominant, so the monster has two small eyes.

Trait	**Allele 1 (heads)**	**Allele 2 (tails)**
Eyes	Two small eyes (E)	One large eye (e)
Eye color	Red (R)	White (r)
Skin color	Green (G)	Blue (g)

Table 2: finish the male monster (row 1 is done for you; the other traits are on your Monster Genetics sheet)

Trait	**Genotype**	**Phenotype**
Eyes	ee	One large eye (done for you)
Eye color		White
Skin color	Gg	
Tail shape		Straight
Tail color	Pp	

UNIT 2.2 · DAY 6 · GLUE IN · WRITE ON IT

Punnett square: skin color and eyes

Skin color: Gg × Gg (together). Then Eyes on your own.

	G	g
G	GG	Gg
g	Gg	gg

Eyes: ___ × ___

UNIT 2.2 · DAY 6 · GLUE IN · REFERENCE

Prediction vs. class count

Baby's skin color	Punnett prediction	Class count (sample)	Class percent
Green	75%	22 of 30	73%
Blue	25%	8 of 30	27%

UNIT 2.2 · DAY 7 · GLUE IN · REFERENCE

UV light and mutations

Hours of UV light	Mutations per 1,000 skin cells
0 (none)	2
1	9
2	21
4	40

UNIT 2.2 · DAY 7 · GLUE IN · WRITE ON IT

Who passes it on? Five cases

Could the next generation inherit it? Write Yes or No.

Case	Where the DNA changed	Inherited?
A	UV hits a skin cell	No (done)
B	Copy error in a sperm-making cell	
C	Copy error in a muscle cell	
D	X-rays hit an egg-making cell	
E	Smoke chemical hits a lung cell	

UNIT 2.2 · DAY 8 · GLUE IN · REFERENCE

Case data: three cases

Case 1: puppy litter

	Alleles for fur color (B = black, b = brown)
Mom	B and b
Dad	B and b
Puppies	4 black (BB or Bb), 2 brown (bb)

Case 2: DNA of a puppy and its parents

	Allele 1	Allele 2
Mom	CAT	CAT
Dad	CAT	CGT
Puppy	CAT	CCT

Case 3: X-rays and mice

Group of mice	X-ray dose	New mutations per 1,000 sperm cells
A	none	3
B	low	12
C	high	25

UNIT 2.2 · DAY 9 · GLUE IN · REFERENCE

The new coat pattern: two kennels

Kennel	Puppies checked	Puppies with a new coat pattern (in neither parent)
A (chemical in the water)	1,000	14
B (clean water)	1,000	3

UNIT 2.2 · DAY 10 · GLUE IN · WRITE ON IT

My target tracker

Write your CFA score out of 5. Below a B (fewer than 4 of 5)? Write Yes under Review. At the end of class, rate each target 1 to 4 (4 = I could teach it).

Target	CFA score (of 5)	Review?	Rating (1 to 4)
LS3-2.1 three sources			
LS3-2.2 sex cells and fertilization			
LS3-2.3 copying errors and UV			
LS3-2.4 read data, make a claim			

UNIT 2.2 · DAY 10 · GLUE IN · REFERENCE

Hydrangea soil data

Soil pH	Flower color (same DNA)
5.0	Blue
6.0	Purple
7.0	Pink